Advance Python Assignment

Topic-6: Class and Object (OOP Concepts)

Q1. Understanding the concepts of classes, objects, attributes, and methods in Python.

Ans.
In Python, a class is a blueprint used to create objects. It defines properties and behaviors that an object can have. An object is an instance of a class and represents a real-world entity. Attributes are variables that store data inside a class, while methods are functions that define the behavior of the class. Classes and objects help in organizing code and make it reusable. This concept is part of object-oriented programming and is widely used in Python for building structured programs.

Example:
```
class Student:
    def __init__(self, name):
        self.name = name

    def show(self):
        print("Name:", self.name)

s1 = Student("Anshu")
s1.show()
```

Q2. Difference between local and global variables in Python.

Ans.
In Python, local variables are defined inside a function and can be used only within that function. They are not accessible outside the function. Global variables are defined outside all functions and can be accessed throughout the program. Local variables are temporary and exist only during function execution, while global variables remain available for the entire program. Understanding the difference helps in managing data and avoiding errors in large programs.

Example:
```
x = 10  # Global variable
```

```
def show():
    y = 5  # Local variable
    print("Local:", y)
    print("Global:", x)

show()
print("Global outside function:", x)
```